MIL-M-28787/119A(EC)
18 February 1982
SUPERSEDING
MIL-M-28787/119(NAVY)
8 April 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, POWER, DIGITAL, KEY CODE BDH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic power NAND gate module.

1.2 Classification. This specification covers one class of modules

Class I as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN. PLEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.1 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3.5.1 Key code. The module key code shall be BDH

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes

3.5.8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/119-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

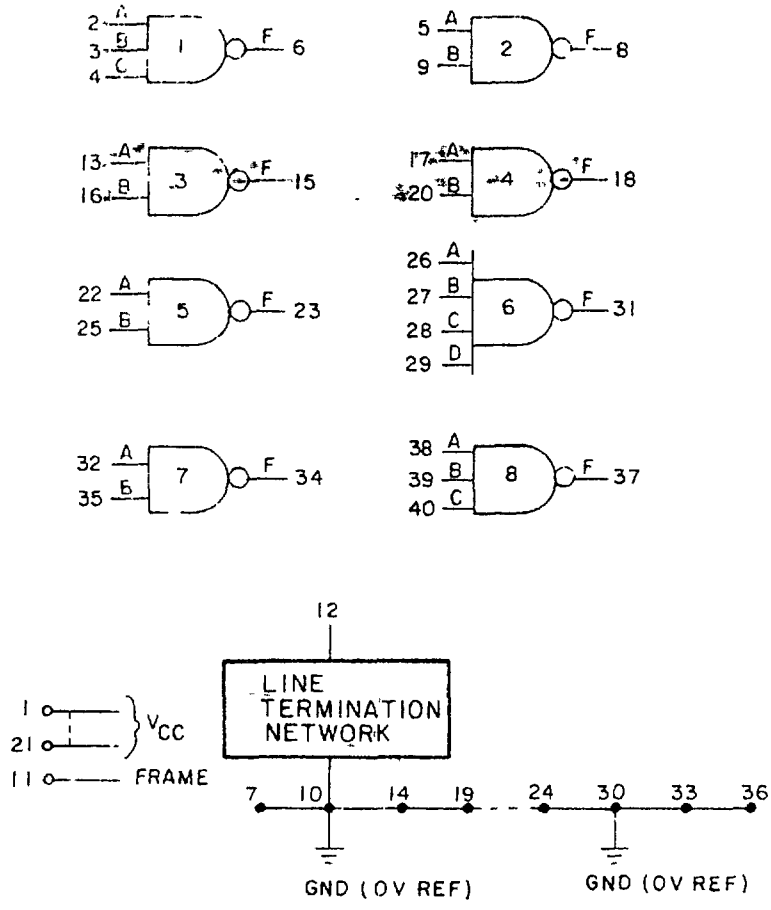
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

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NOTE: Dashed lines indicate connections which are optional within the module

FIGURE 1 LDH function diagram, power NAND gate

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TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A20	Zener applied potential, cond A	III	S	3 8	5.5	3 5	5 5	3 8	5 5	3.5	5.5	V
B1-B8	Zener applied potential, cond B	III	S	0.00	0.40	0 00	0.45	0.00	0 40	0.00	0 45	V
C1-C8	V_{OUT} (low loaded)	IV	S	0 0	1.1	0 0	1 1	0 0	1.1	0.0	1.1	V
D1-D6	Continuity	V	X	0	1	0	1	0	1	0	1	Ω
D7	Continuity	V	S	0	1	0	1	0	1	0	1	Ω
E1,E2	Isolation	V	S	10	∞	10	∞	10	∞	10	∞	MΩ
F1-F20	Output voltage, high level	.	X	3.8	5 5	3.5	5 5	3 8	5 5	3 5	5.5	V
G1-G8	Output voltage, low level	VI	X	0.00	0.40	0.00	0.45	0 00	0.40	0.00	0.45	V
H1-H8	Output short circuit current	VII	X	16	46	16	45	16	46	16	46	mA
J1-J20	Input current, low level	VIII	X	0.2	1.6	0.2	1.6	0 2	1.6	0.2	1.6	mA
K1-K34	Input current, high level	VIII	X	0.0	2.0	0 0	4 0	0 0	3.0	0 0	5 0	µA
L1-L20	Input capacitance	IX	Q	2	16	2	16	2	16	2	16	pF
M1	Power supply current, high level output	X	S	0.0	21 8	0 0	21 8	0 0	21.8	0 0	21 8	mA
M2	Power supply current, low level output	X	S	0	133	0	133	0	133	0	133	mA
N1-N8	Output crosstalk, high level	XI	Q	0.0	1.0	0 0	1 0	0.0	1.0	0.0	1.0	V
P1-P8	Output crosstalk, low level	XI	Q	0.0	0.8	0 0	0.8	0.0	0 8	0.0	0.8	V
Q1-Q8	Transition time (t_{THL})	XII	S	6	40	6	40	6	40	6	40	ns
Q9-Q16	Transition time (t_{TLH})	XII	S	35	130	35	130	35	130	35	130	ns
R1-R8	Propagation delay time (t_{PHL})	XIII	S	15	75	15	75	15	75	15	75	ns
R9-R16	Propagation delay time (t_{PLH})	XIII	S	15	55	15	55	15	55	15	55	ns
S1-S8	Output pulse amplitude, high level	XIV	S	3.8	5.5	3.5	5.5	3 8	5.5	3.5	5 5	V

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S9-S16	Output pulse amplitude, low level	XIV	S	0.00	0.50	0 00	0 50	0 00	0.50	0.00	0.50	V
T1	10 MHz terminator amplitude	XV	X	0 940	1 220	0 924	1.236	0.930	1.230	0.918	1.242	Vp-p
T2	1 MHz terminator amplitude	XV	X	1 786	1 974	1.772	1.988	1.777	1.983	1 762	1.998	Vp-p
U	Power supply transient rise time	XVI	S	10	140	10	140	10	140	10	140	µs
V1-V3	Parallel operation output voltage, high level	XVII	S	3.8	5.5	3 5	5 5	3.8	5 5	3.5	5.5	V
W1-W3	Parallel operation output voltage, low level	XVII	S	0.00	0 40	0 00	0.45	0.00	0.40	0.00	0.45	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions D1 through D6, F through K and M through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests D1 through D6, F through K and M through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D1 through D6 and F through W shall be performed in the order specified in 4 4 2 using the 25° ±5°C ambient end-of-life requirements of table I

5. PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification M28787/119 supersedes Department of the Navy, Naval Ordnance System Command Drawing No. 10001-2658715 D dated 2 February 1971. Part number M28787/119-1 supersedes part numbers 2658715 and M28787/119 These parts are interchangeable

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits of table I.

Maximum power dissipation	
All outputs at low level	730 milliwatts (outputs not loaded)
All outputs at high level	1120 milliwatts (outputs not loaded)
Speed	DC to 10.8 MHz
Input current	
Low level	1.6 mA maximum
High level	5.0 µA maximum
Input threshold	
Low level	0.0 V minimum, 1.2 V maximum
High level	1.8 V minimum, V_{CC} maximum
Output voltage	
Low level	0.0 V minimum, 0.45 V maximum
High level	3.5 V minimum, 5.5 V maximum
Output drive current (maximum capability)	
Low level	35.2 mA
High level	-220 µA

6.5 Module function. The module consists of five 2-input power NAND gates, two 3-input power NAND gates, one 4-input power NAND gate, and one line terminating network.

6.5.1 Input to output relationship. Each gate has the input to output relationship for positive logic shown in table II.

6.5 2 Parallel operation. Power NAND gates are capable of being connected for parallel operation (commoned inputs and commoned outputs). The maximum permissible number of parallel NAND gates is three. Wired OR connections are not permitted (commoned outputs and non-commoned inputs).

6 5.3 Line termination network. The line termination network consists of a resistor and capacitor network with an internal zero volt return to circuit ground This network provides a means of terminating a transmission line to minimize ringing on fast rise time pulses.

6.5 4 Module speed. When driving twenty-two standard logic loads (SLL) the module operates from DC to 0.8 MHz (50 percent duty cycle). The high level and low level output voltage levels correspond to the limits shown in Table I. When driving one SLL, the module operates from DC to 10 MHz (50 percent duty cycle). The module high level output voltage is > 2.0 V and the low level output voltage is < 0.50 V. NOTE. Capacitive loading of 10 pF represents nominal module loading and does not take into account Type II wiring and termination capacitance.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity:
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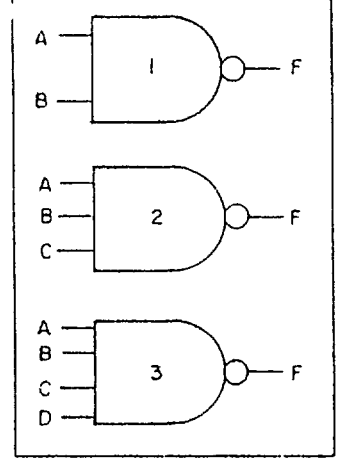
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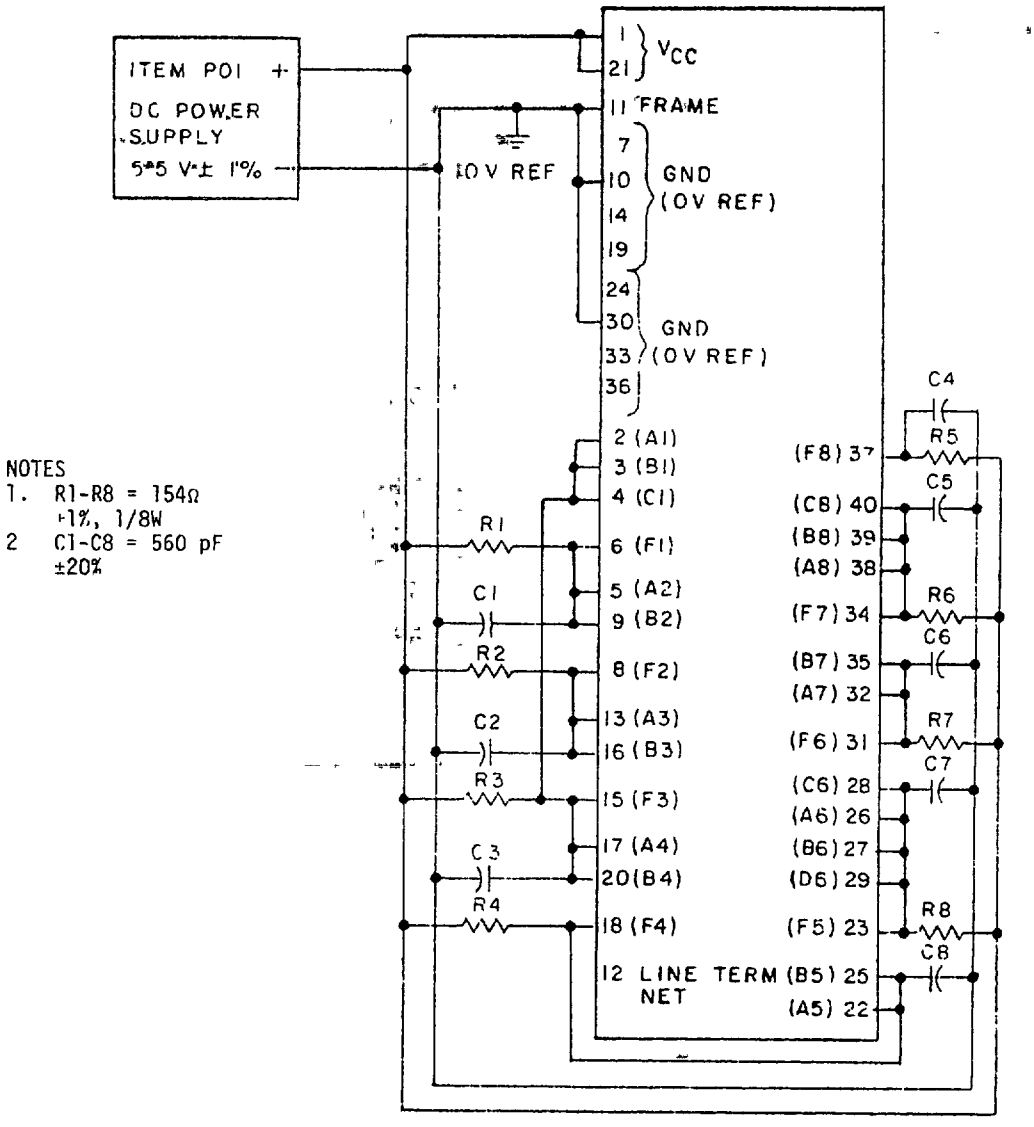
TABLE I. Input to output relationship for each NAND gate

Inputs				Output
A	B	C	D	F
L	X	X	X	H
X	L	X	X	H
X	X	L	X	H
X	X	X	L	H
H	H	H	H	L



1/ A, B, C, D = Gate inputs
2/ F = Gate outputs
3/ L = Low level
4/ H = High level
5/ X = Either high or low level

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NOTES
1. R1-R8 = 154Ω ±1%, 1/8W
2 C1-C8 = 560 pF ±20%

FIGURE 2 Operating temperature and life test circuit

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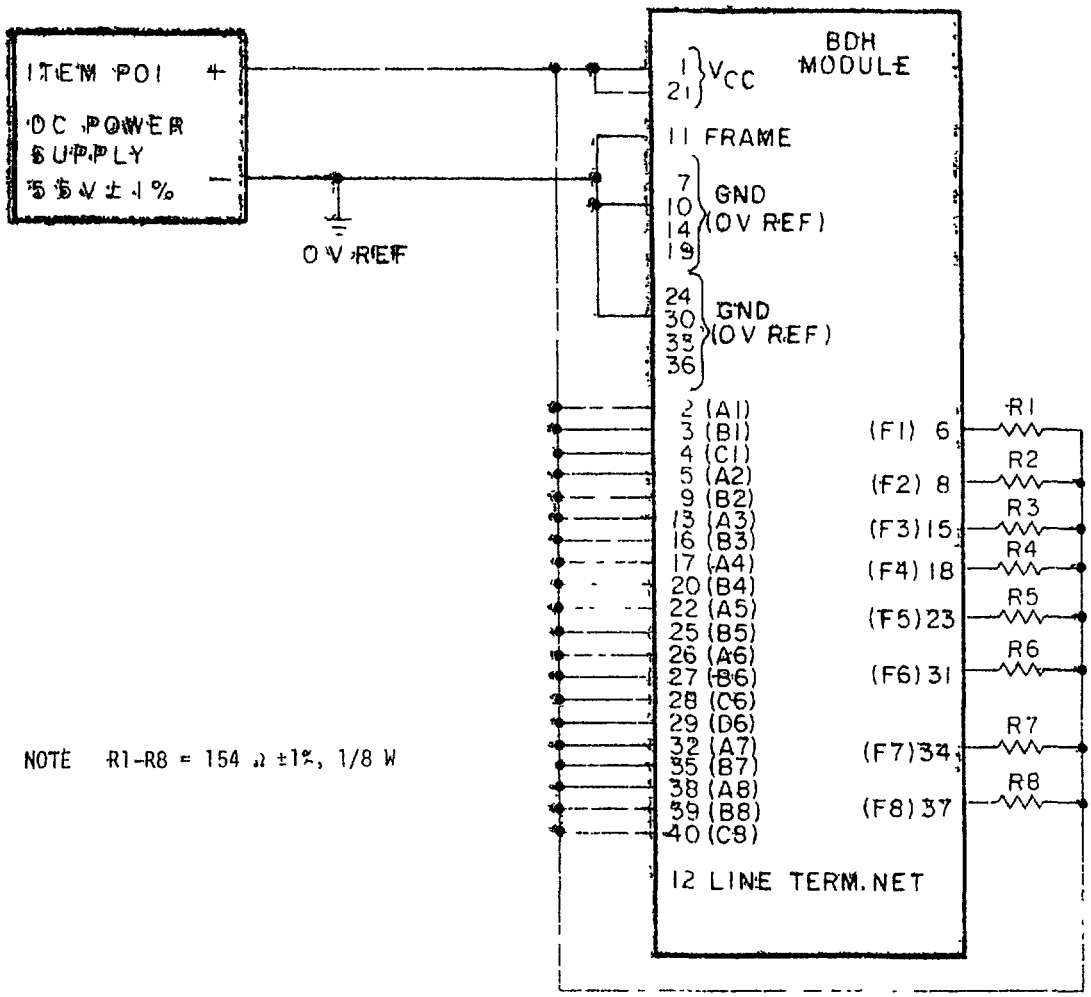


FIGURE 3 Component temperature test circuit

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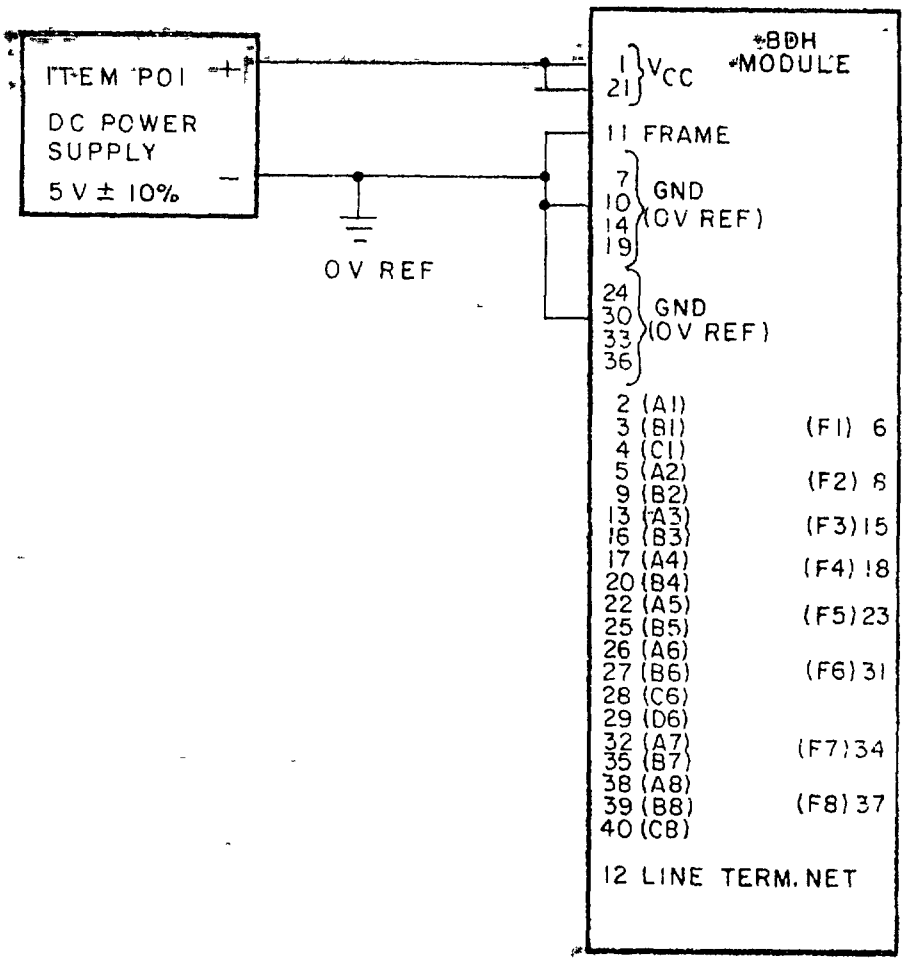


FIGURE 4 Vibration test circuit

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TABLE III. Zener applied potential test

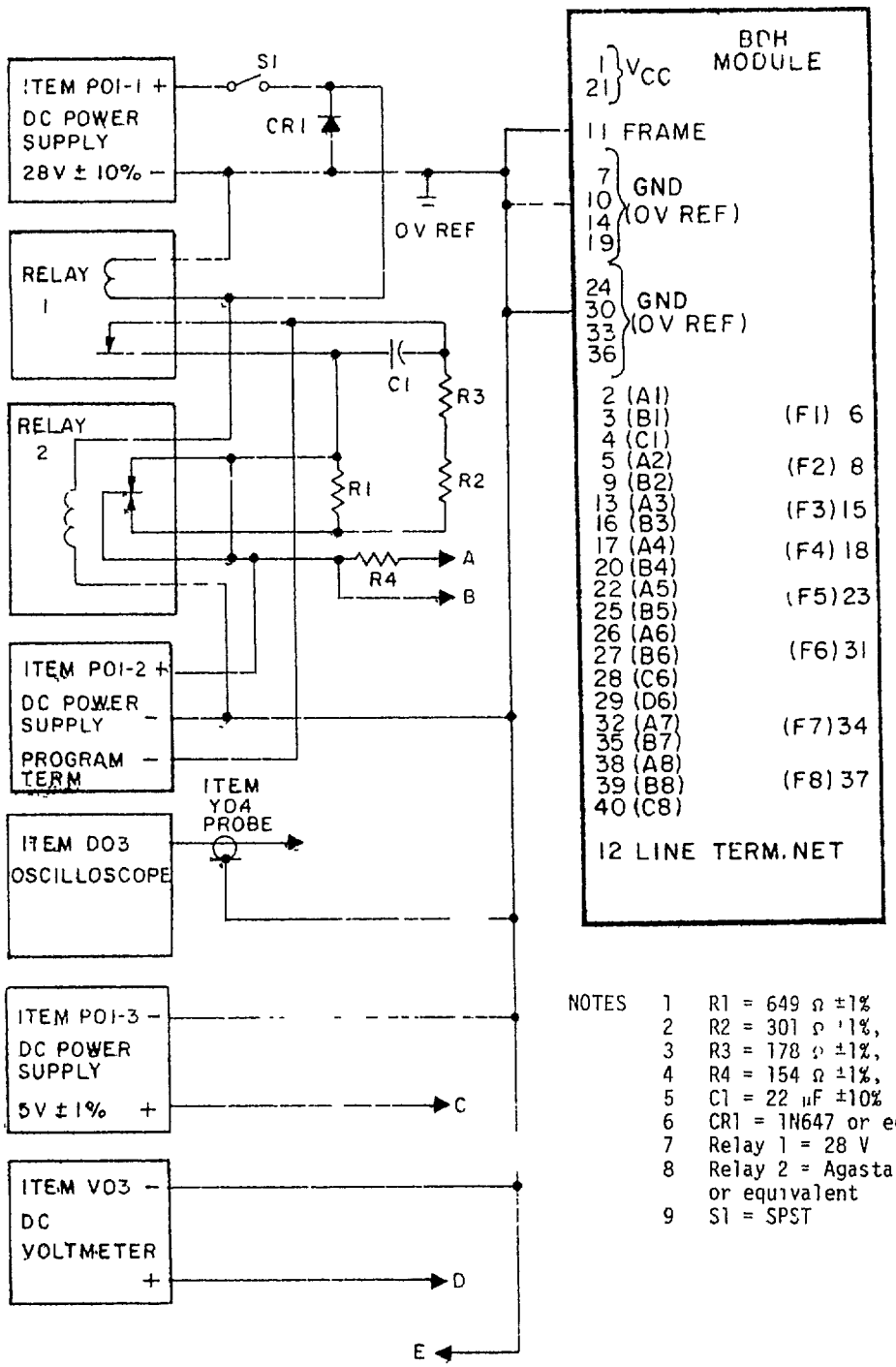
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
A1	A1 Zener applied potential, cond A	2	1, 21	Open	6	3, 4	3.8	5.5	V	5
A2	B1	3			6	2, 4				
A3	C1	4			6	2, 3				
A4	A2	5			8	9				
A5	B2	9			8	5				
A6	A3	13			15	16				
A7	B3	16			15	13				
A8	A4	17			18	20				
A9	B4	20			18	17				
A10	A5	22			23	25				
A11	B5	25			23	22				
A12	A6	26			31	27,28,29				
A13	B6	27			31	26,28,29				
A14	C6	28			31	26,27,29				
A15	D6	29			31	26,27,28				
A16	A7	32			34	35				
A17	B7	35			34	32				
A18	A8	38			37	39, 40				
A19	B8	39			37	38, 40				
A20	C8	40			37	38, 39				

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TABLE III. Zener applied potential test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
B1	A1 Zener applied potential, cond B	2	Open	1, 21	6	Open	0.00	0.40	V	5
B2	A2	5			8					
B3	A3	13			15					
B4	A4	17			18					
B5	A5	22			23					
B6	D6	26			31					
B7	A7	32			34					
B8	A8	38			37					

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NOTES
1. R1 = 649 Ω ±1% 1/8 W
2. R2 = 301 Ω ±1%, 1/8 W
3. R3 = 178 Ω ±1%, 1/8 W
4. R4 = 154 Ω ±1%, 1/8 W
5. C1 = 22 µF ±10%
6. CR1 = 1N647 or equivalent
7. Relay 1 = 28 V
8. Relay 2 = Agastat Model 2112 or equivalent
9. S1 = SPST

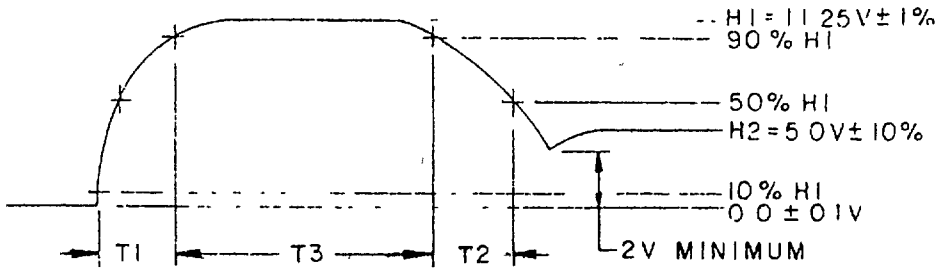
FIGURE 5 Zener applied potential test circuit for table III

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NOTES

10 Resistors R1, R2, R3 and capacitor C1 are programming components for item P01-2 and are based on a programming constant of 100 ohms per volt

11. Using item D03 with item Y04 probe, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay relay after switch S1 is closed



Waveform parameters

T1 = 100 $^{+100}_{-50}$ ms

T2 = 100 $^{+400}_{-50}$ ms

T3 = 500 ms ±10%

12 For each test condition, switch S1 shall be closed and voltage measurements shall be made after a minimum of 2 seconds after switch S1 is closed At the conclusion of each test condition, switch S1 shall be opened

13 The circuit is drawn showing item P01-2 with a negative reference terminal (non-commoned programming terminal labeled (-)). The figure, as shown, will not operate properly if item P01-2 has a positive non-commoned programming reference terminal NOTE Load current must not pass through the sensing network

FIGURE 5 Zener applied potential test circuit for table III - Continued

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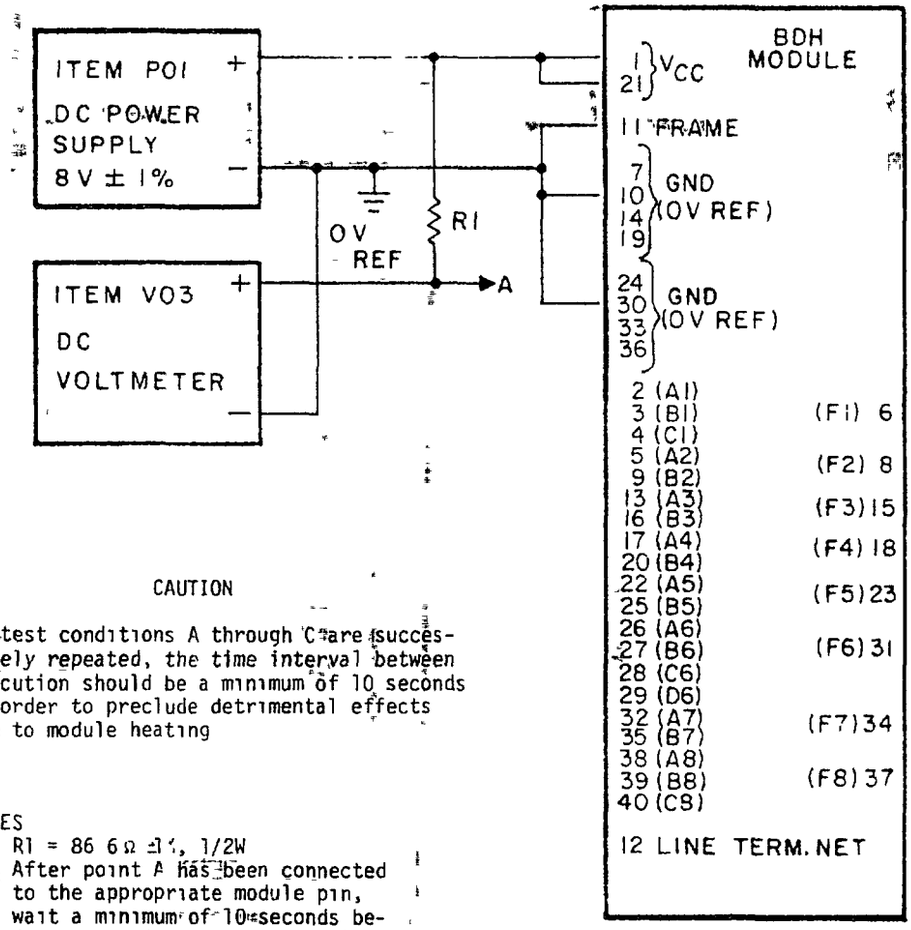
TABLE IV. V_{OUT} (low loaded) test

Test condition	Parameter	Connect A to pin	Initial limits 25° +5°C ambient Min	Initial limits 25° +5°C ambient Max	Unit	Test figure
C1	F1 V_{OUT} (low loaded)	6	0.0	1.1	V	6
C2	F2	8				
C3	F3	15				
C4	F4	18				
C5	F5	23				
C6	F6	31				
C7	F7	34				
C8	F8	37				

TABLE V. Continuity and isolation test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Continuity	Open	Open	10	7	0	1	Ω	7
D2				10	14				
D3				10	19				
D4				30	24				
D5				30	33				
D6				30	36				
D7				11	Frame				
E1	Isolation	Frame	All other pins tied together	Open	Open	10	∞	MΩ	
E2	Isolation	Open	Open	12	All other pins tied together	10	∞	MΩ	

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CAUTION

If test conditions A through C are successively repeated, the time interval between execution should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating

NOTES
1. R1 = 86.6 Ω ±1%, 1/2W
2. After point A has been connected to the appropriate module pin, wait a minimum of 10 seconds before measuring the output voltage.

FIGURE 6 V_{OUT} (low-loaded) test circuit for table IV.

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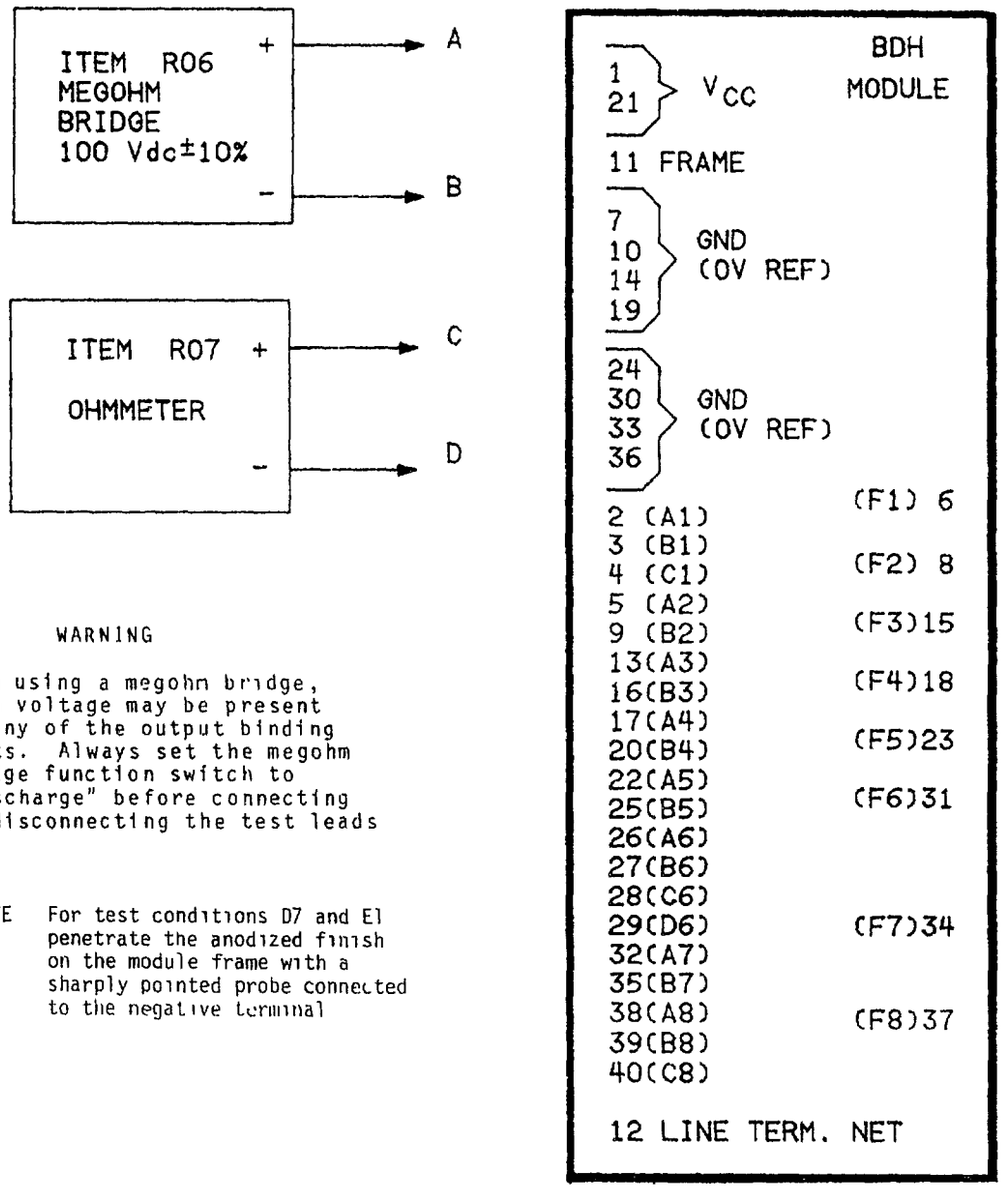


FIGURE 7 Continuity and isolation test circuit for table V

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TABLE VI Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	F1 Output voltage, high level	2	Open	6	6	3.8	5.5	V	8
F2	F1	3		6	6				
F3	F1	4		6	6				
F4	F2	5		8	8				
F5	F2	9		8	8				
F6	F3	13		15	15				
F7	F3	16		15	15				
F8	F4	17		18	18				
F9	F4	20		18	18				
F10	F5	22		23	23				
F11	F5	25		23	23				
F12	F6	26		31	31				
F13	F6	27		31	31				
F14	F6	28		31	31				
F15	F6	29		31	31				
F16	F7	32		34	34				
F17	F7	35		34	34				
F18	F8	36		37	37				
F19	F8	39		37	37				
F20	F8	40		37	37				

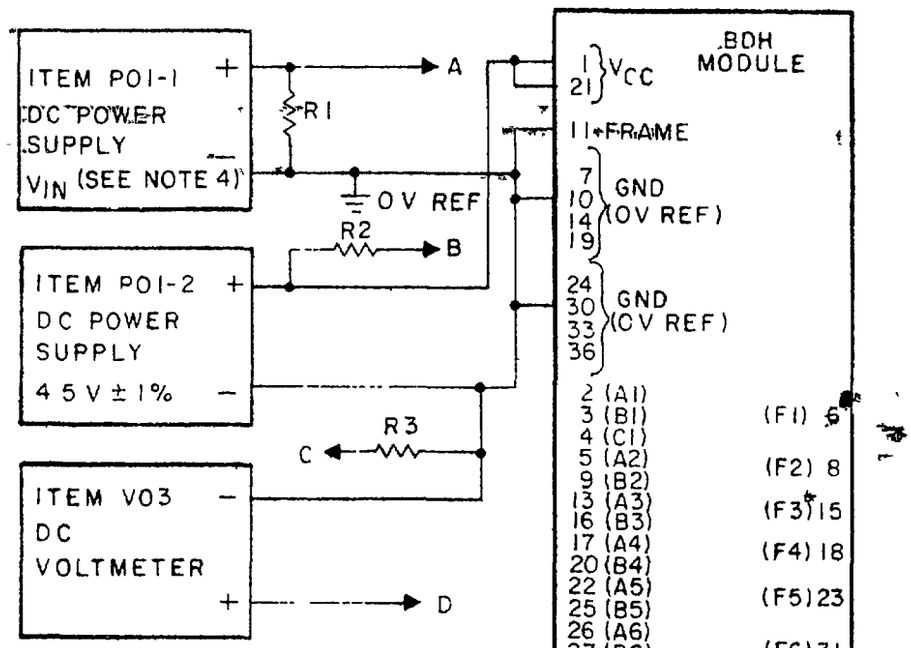
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TABLE VI. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	F1 Output voltage, low level	2, 3, 4	6	Open	6	0.00	0.40	V	8
S2	F2	5, 9	8		8				
S3	F3	13, [illegible]	15		15				
S4	F4	17, [illegible]	18		18				
S5	F5	22, [illegible]	23		23				
S6	F6	26, [illegible] 28, [illegible]	31		31				
S7	F7	32, 35	34		34				
S8	F8	38, [illegible], 40	37		37				

MIL-M-28787/119A(EC)

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NOTES
1 R1 = 53.6Ω ±1%, 1/2W
2 R2 = 115Ω ±1%, 1/2W
3 R3 = 16.9 kΩ ±1%, 1/8W
4 Use appropriate V_{IN} for specified test condition and temperature listed below

Test condition	Temperature (°C)	V_{IN} (volts)
F1-F20	0	1 20
	25	1 10
	60	0 95
	80	0 80
G1-G8	0	2 0
	25	1 9
	60	1.8
	80	1 8

0°, 60°, and 80°C refer to operating temperatures (module fin), 25°C refers to ambient temperature For 80°C use the 0° to 60°C end-of-life limits "

FIGURE 8 Output voltage (high-level and low-level) test circuit for table VI

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TABLE VII Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	F1 Ouput short circuit current	Oper	6	2, 3, 4	16	46	mA	9
H2	F2		8	5, 9				
H3	F3		15	13, 16				
H4	F4		18	17, 20				
H5	F5		23	22, 25				
H6	F6		31	26, 27 28, 29				
H7	F7		34	32, 35				
H8	F8		37	38, 39 40				

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TABLE VIII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	A1 Input current, low level	Open	2	Open	0.2	1.6	mA	9
J2	B1		3					
J3	C1		4					
J4	A2		5					
J5	B2		9					
J6	A3		13					
J7	B3		16					
J8	A4		17					
J9	B4		20					
J10	A5		22					
J11	B5		25					
J12	A6		26					
J13	B6		27					
J14	C6		28					
J15	D6		29					
J16	A7		32					
J17	B7		35					
J18	A8		38					
J19	B8		39					
J20	C8		40					

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TABLE VIII. Input current (low-level and high-level) test - Continued.

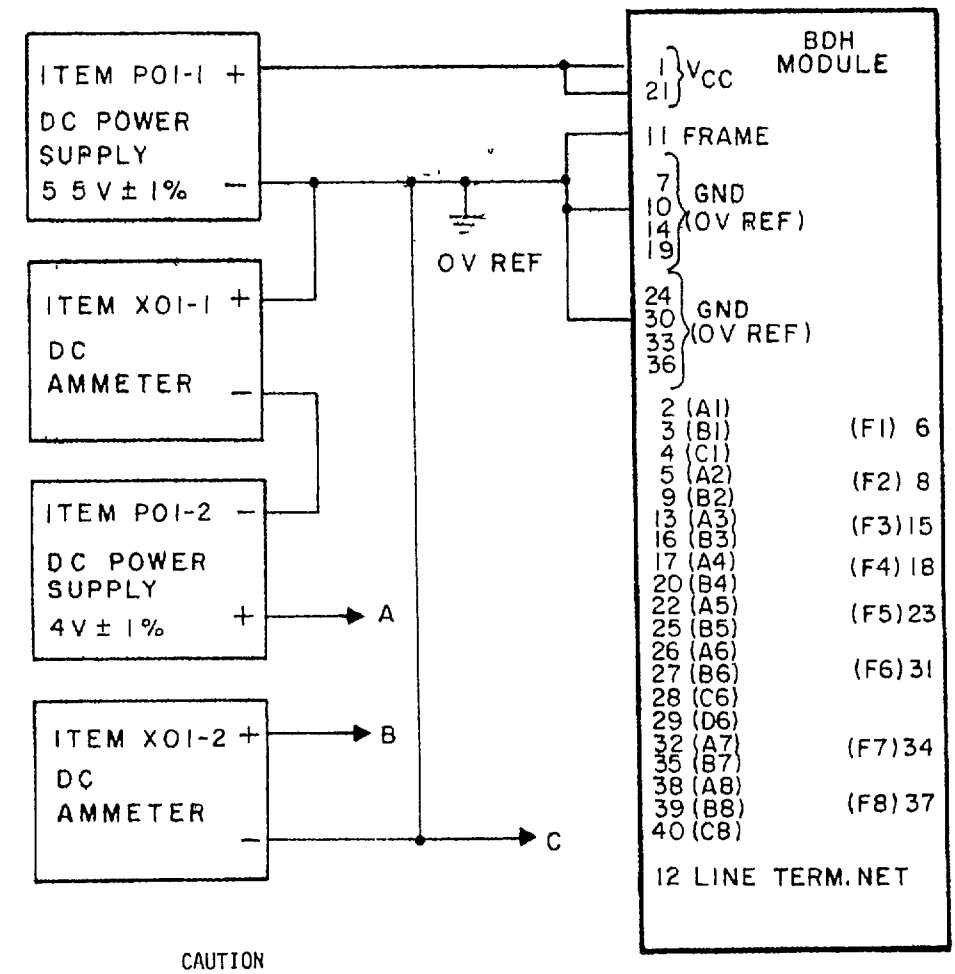
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	A1 Input current, high level	2	Open	3	0.0	2.0	µA	9
K2	A1	2		4				
K3	B1	3		2				
K4	B1	3		4				
K5	C1	4		2				
K6	C1	4		3				
K7	A2	5		9				
K8	B2	9		5				
K9	A3	13		16				
K10	B3	16		13				
K11	A4	17		20				
K12	B4	20		17				
K13	A5	22		25				
K14	B5	25		22				
K15	A6	26		27				
K16	A6	26		28				
K17	A6	26		29				
K18	B6	27		26				
K19	B6	27		28				
K20	B6	27		29				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K21	C6 Input current, high level	28	Open	26	0.0	2.0	µA	9
K22	C6	28		27				
K23	C6	28		29				
K24	D6	29		26				
K25	D6	29		27				
K26	D6	29		28				
K27	A7	32		35				
K28	B7	35		32				
K29	A8	38		39				
K30	A8	38		40				
K31	B8	39		38				
K32	B8	39		40				
K33	C8	40		38				
K34	C8	40		39				

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CAUTION

Do not isolate chassis of ammeter X01 from earth ground.

NOTE Since ammeters X01-1 and X01-2 are not used simultaneously, one ammeter X01 is required.

CAUTION

The chassis of the item P01-2 shall be isolated from earth ground.

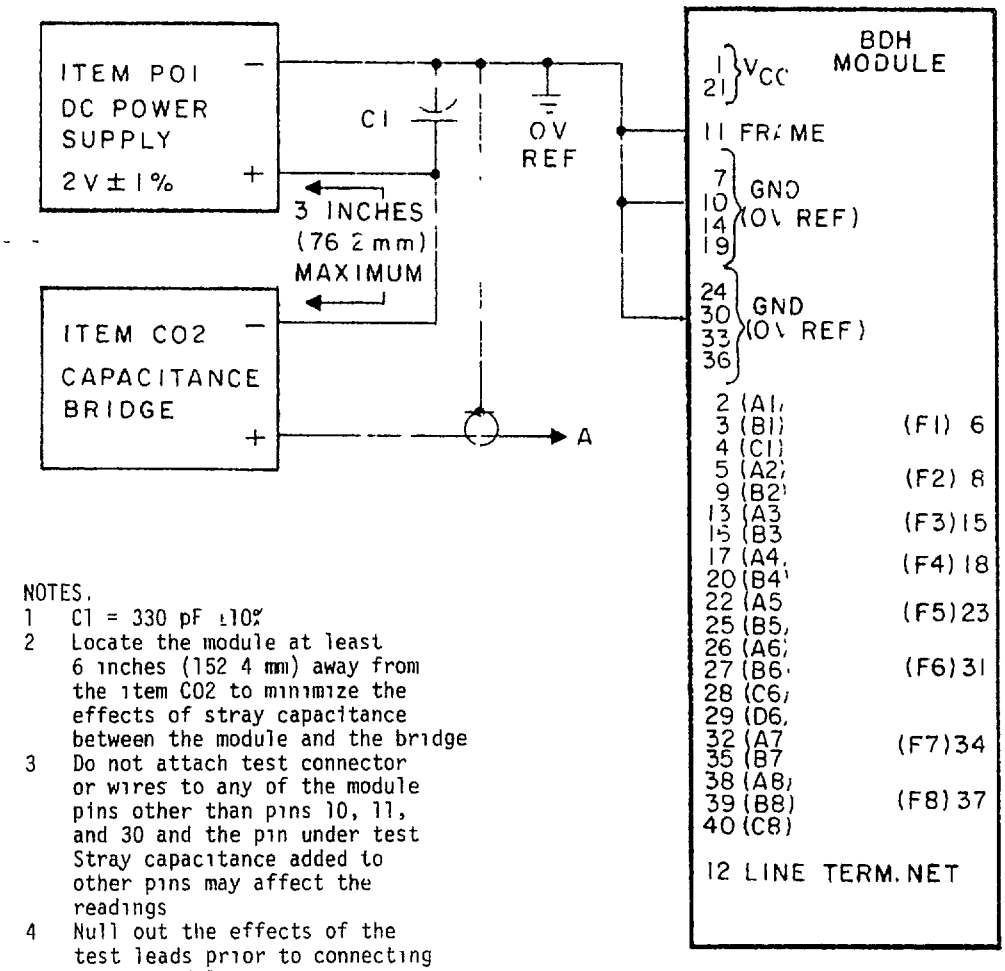
FIGURE 9 Output short circuit current and input current (low-level and high-level) test circuit for tables VII and VIII.

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TABLE IX. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
L1	A1 Input capacitance	2	2	16	pF	10
L2	B1	3				
L3	C1	4				
L4	A2	5				
L5	B2	9				
L6	A3	13				
L7	B3	16				
L8	A4	17				
L9	B4	20				
L10	A5	22				
L11	B5	25				
L12	A6	26				
L13	B6	27				
L14	C6	28				
L15	D6	29				
L16	A7	32				
L17	B7	35				
L18	A8	38				
L19	B8	39				
L20	C8	40				

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NOTES.
1 C1 = 330 pF ±10%
2 Locate the module at least 6 inches (152 4 mm) away from the item CO2 to minimize the effects of stray capacitance between the module and the bridge
3 Do not attach test connector or wires to any of the module pins other than pins 10, 11, and 30 and the pin under test Stray capacitance added to other pins may affect the readings
4 Null out the effects of the test leads prior to connecting to the module
5 Capacitance bridge (CO2) setup
Frequency = 1 MHz ±5%.
V_{pk} = 0 5 Vrms maximum.

FIGURE 10 Input capacitance test circuit for table IX

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TABLE A. Power supply current (high-level and low-level) output test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient, Min	Initial limits 25° ±5°C ambient, Max	Unit	Test figure
M1	Power supply current, high level output	10	0 0	21.8	mA	11
M2	Power supply current, low level output	1	0	133	mA	11

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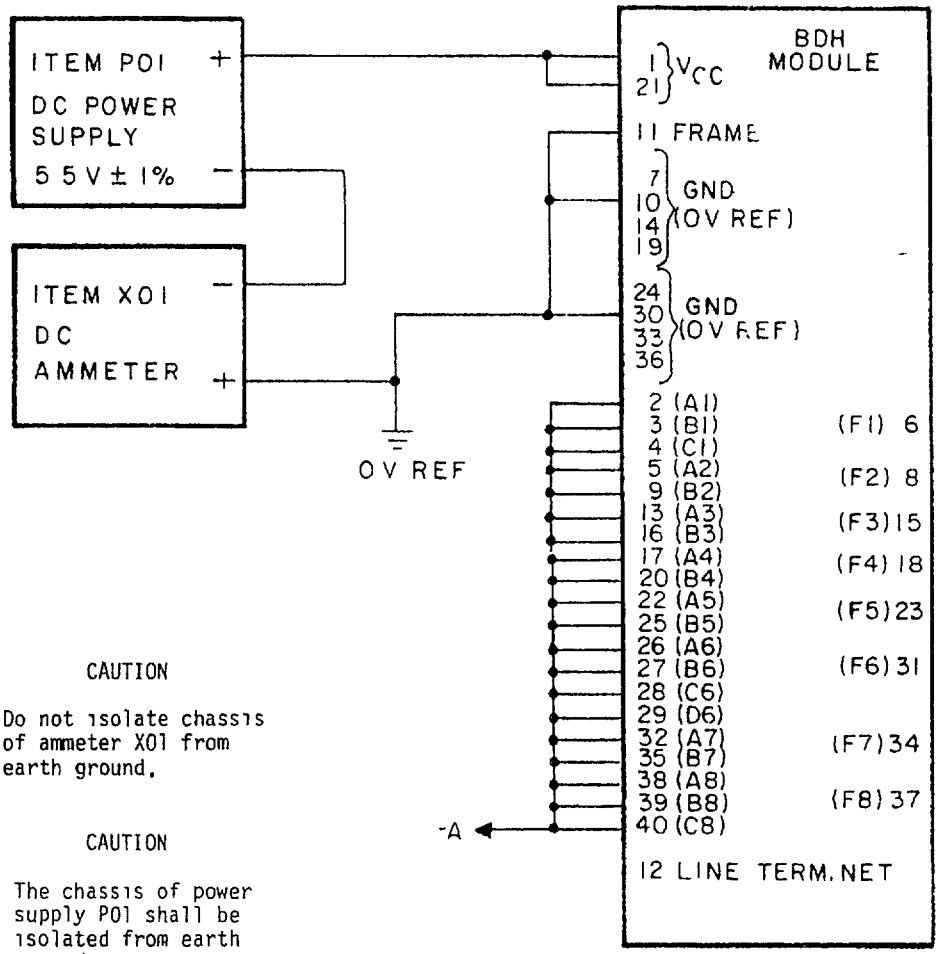


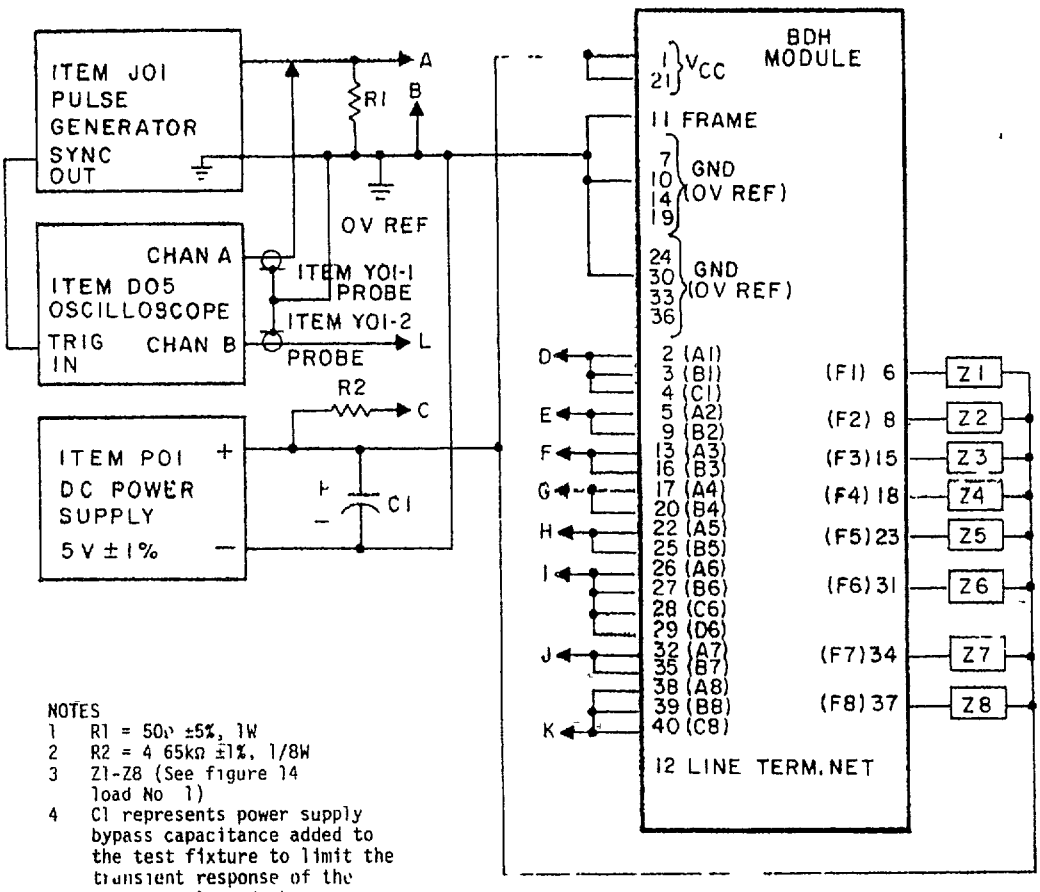
FIGURE 11 Power supply current (high-level and low-level) output test circuit for table X

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TABLE VI Output crosstalk (high-level and low-level) test

Test condition	Parameter	Connect				Initial limits 25° ±5°C & ambient		Unit	Test figure
		A to pin	B to pin	C to pin	L to pin	Min	Max		
N1	F1 Output crosstalk, high level	E,F,G,H, I,J,K	D	Open	6	0.0	1.0	V	12,13
N2	F2	D,F,G,H, I,J,K	E		8				
N3	F3	D,E,G,H, I,J,K	F		15				
N4	F4	D,E,F,H, I,J,K	G		18				
N5	F5	D,E,F,G, I,J,K	H		23				
N6	F6	D,E,F,G, H,J,K	I		31				
N7	F7	D,E,F,G, H,I,K	J		34				
N8	F8	D,E,F,G, H,I,J	K		37				
P1	F1 Output crosstalk, low level	E,F,G,H, I,J,K	Open	D	6		0.8		
P2	F2	D,F,G,H, I,J,K		E	8				
P3	F3	D,E,G,H, I,J,K		F	15				
P4	F4	D,E,F,H, I,J,K		G	18				
P5	F5	D,E,F,G, I,J,K		H	23				
P6	F6	D,E,F,G, H,J,K		I	31				
P7	F7	D,E,F,G, H,I,K		J	34				
P8	F8	D,E,F,G, H,I,J		K	37				

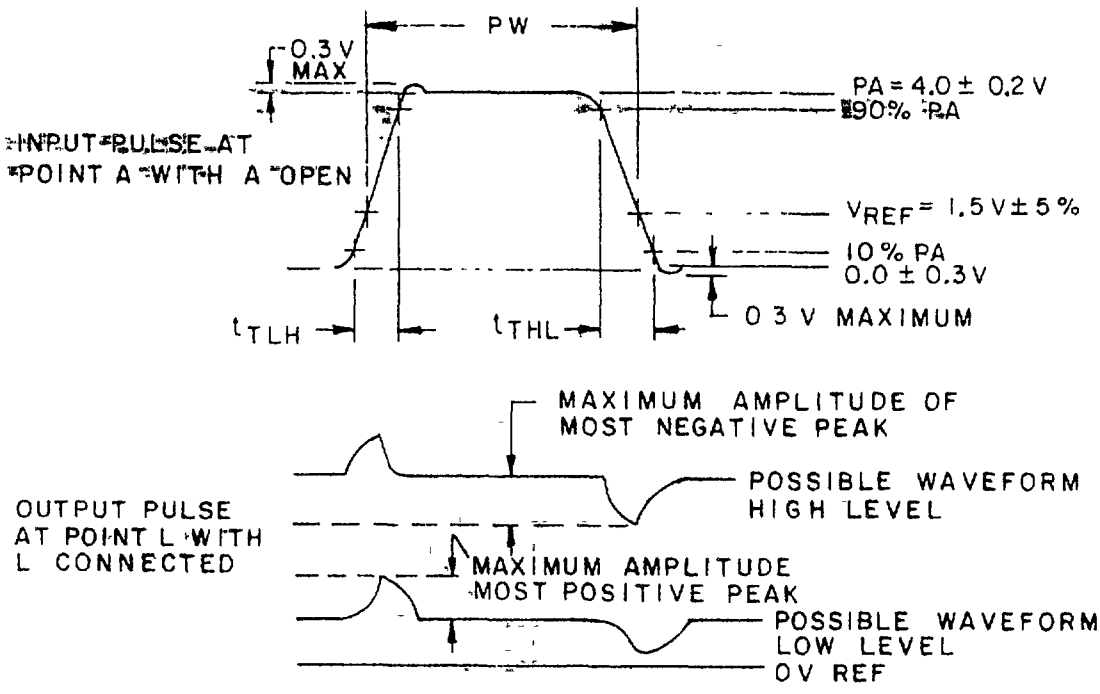
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NOTES
1. R1 = 50Ω ±5%, 1W
2. R2 = 4.65kΩ ±1%, 1/8W
3. Z1-Z8 (See figure 14 load No. 1)
4. C1 represents power supply bypass capacitance added to the test fixture to limit the transient response of the power supply and the power supply transmission paths to module current demands. Capacitance may be added as required.

FIGURE 12. Output crosstalk (high-level and low-level) test circuit for table XI.

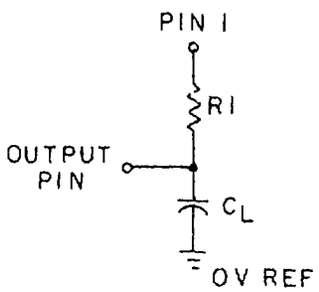
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NOTE.
1 Input pulse parameters:
PW = 0 9 +0 1 µs.
$t_{TLH} = t_{THL}$ = 20 +1ns
PRR = 100 kHz ±10

FIGURE 13 Output crosstalk (high-level and low-level) test circuit waveforms for table XI

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LOAD NO. 1

NOTES
1 For test conditions N1-N8 and P1-P8
(a) R1 = 154Ω ±1%, 1/8W
(b) C_L = 30 ±5 pF (includes stray, probe, and compensating capacitor, as required)
(c) With the module unplugged, the oscilloscope probe disconnected, and the 154 ohm load resistors disconnected from their respective output pins, use item C02 to measure the capacitance between each test connector output pin and pins 10, 11, and 30 combined The resistors shall be disconnected from their respective output pins as close to their bodies as pratical The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin The capacitance shall be measured at 1 MHz ±10% Reconnect the 154 ohm resistors prior to using the test connector for module testing
(d) The loop length from pin 1 to each load resistor and output, including the length of the resistor body, shall not exceed 4 5 inches (114 3 mm)

LOAD NO 2

2 For test conditions Q1-Q8 and R1-R8
(a) Same as 1(a).
(b) C_L = 500 ±25 pF (includes stray, probe, and compensating capacitor, as required)
(c) Same as 1(c), except use item C03 set at 10 kHz ±10%
(d) Same as 1(d)

FIGURE 14 BDH logic load circuit

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TABLE XII. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	F1 Transition time (t_{THL})	2	6	3,4	6	40	ns	15,16
Q2	F2	5	8	9				
Q3	F3	13	15	16				
Q4	F4	20	18	17				
Q5	F5	25	23	22				
Q6	F6	26	31	27,28,29				
Q7	F7	32	34	35				
Q8	F8	40	37	39,38				
Q9	F1 Transition time (t_{TLH})	2	6	3,4	35	130		
Q10	F2	5	8	9				
Q11	F3	13	15	16				
Q12	F4	20	18	17				
Q13	F5	25	23	22				
Q14	F6	26	31	27,28,29				
Q15	F7	32	34	35				
Q16	F8	40	37	39,38				

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TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

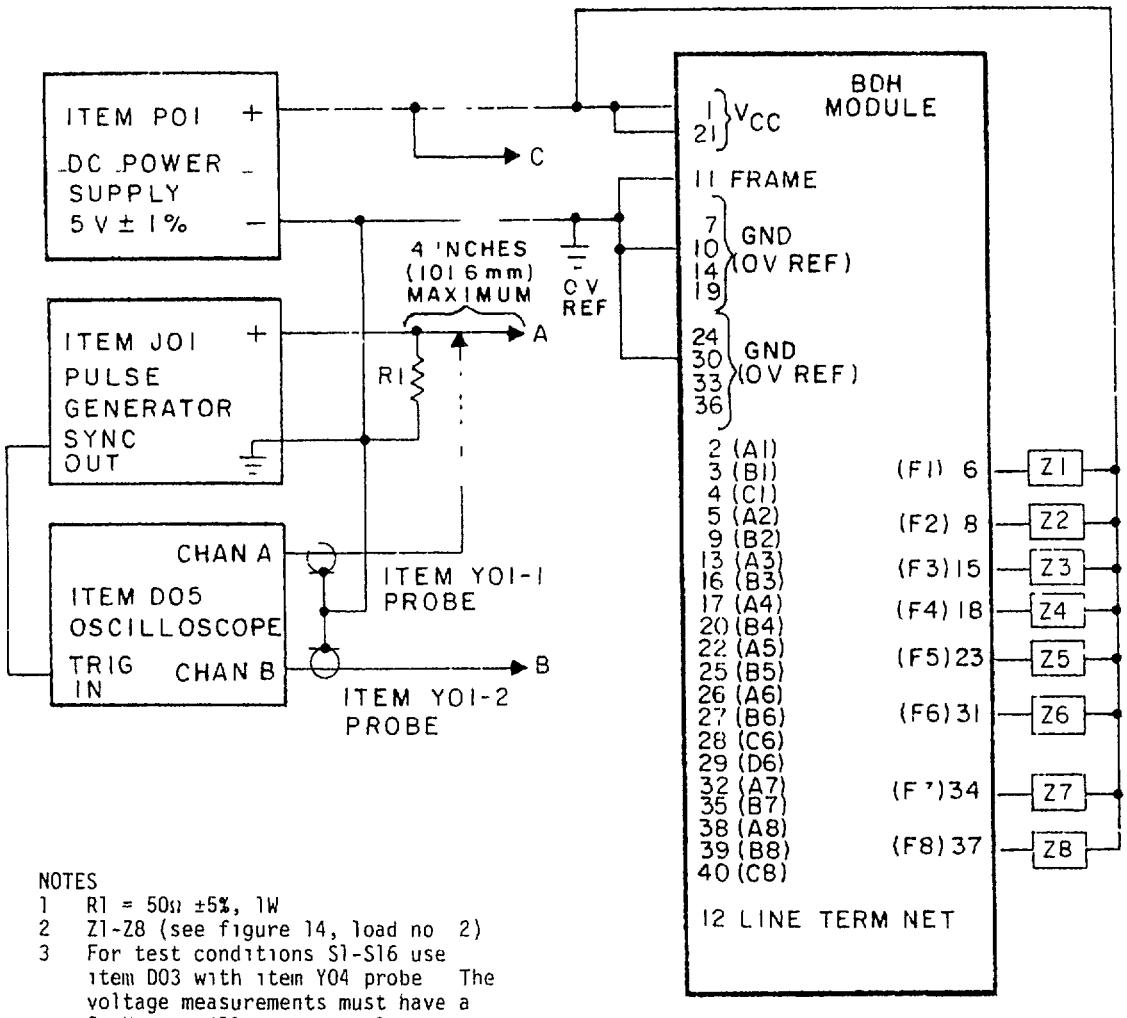
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	F1 Propagation delay time (t_{PHL})	2	6	3,4	15	75	ns	15,16
R2	F2	5	8	9				
R3	F3	13	15	16				
R4	F4	20	18	17				
R5	F5	25	23	22				
R6	F6	26	31	27,28,29				
R7	F7	32	34	35				
R8	F8	40	37	39,38				
R9	F1 Propagation delay time (t_{PLH})	2	6	3,4		55		
R10	F2	5	8	9				
R11	F3	13	15	16				
R12	F4	20	18	17				
R13	F5	25	23	22				
R14	F6	26	31	27,28,29				
R15	F7	32	34	35				
R16	F8	40	37	39,38				

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TABLE XIV. Output pulse amplitude (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	F1 Output pulse amplitude, high level	2	6	3,4	3.8	5.5	V	15,16
S2	F2	5	8	9				
S3	F3	13	15	16				
S4	F4	20	18	17				
S5	F5	25	23	22				
S6	F6	26	31	27,28,29				
S7	F7	32	34	35				
S8	F8	40	37	39,38				
S9	F1 Output pulse amplitude, low level	2	6	3,4	0.00	0.50		
S10	F2	5	8	9				
S11	F3	13	15	16				
S12	F4	20	18	17				
S13	F5	25	23	22				
S14	F6	26	31	27,28,29				
S15	F7	32	34	35				
S16	F8	40	37	39,38				

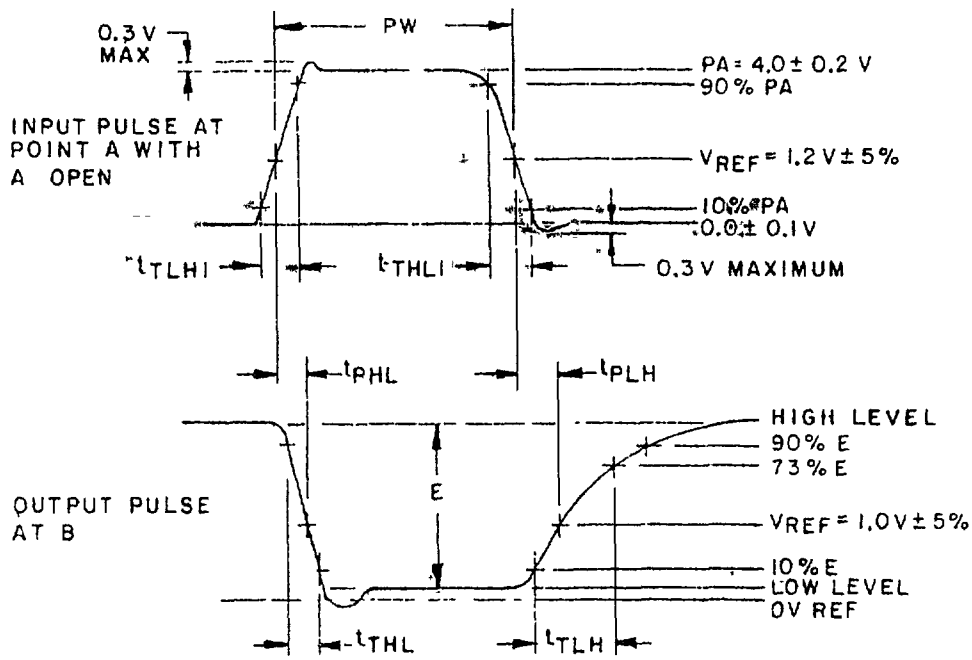
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NOTES
1 R1 = 50Ω ±5%, 1W
2 Z1-Z8 (see figure 14, load no 2)
3 For test conditions S1-S16 use item D03 with item Y04 probe The voltage measurements must have a 5 mV per millimeter resolution

FIGURE 15 Transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PHL} and t_{PLH}) and output pulse amplitude (high-level and low-level) test circuit for table XII, XIII and XIV

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NOTE Input pulse parameters
PW = 1 µs ±10%
t_{TLH1} t_{THL1} = 20 2ns
PRR 100 kHz ±10%

FIGURE 16 Transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PHL} and t_{PLH}) and output pulse amplitude (high-level and low-level) test circuit waveforms for tables XII, XIII and XIV

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TABLE XV. Terminator amplitude (10 MHz and 1 MHz) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		C to pin	Min	Max		
T1	Terminator amplitude, 10 MHz	B	0.940	1.220	Vp-p	17
T2	Terminator amplitude, 1 MHz	B	1 786	1.974	Vp-p	17

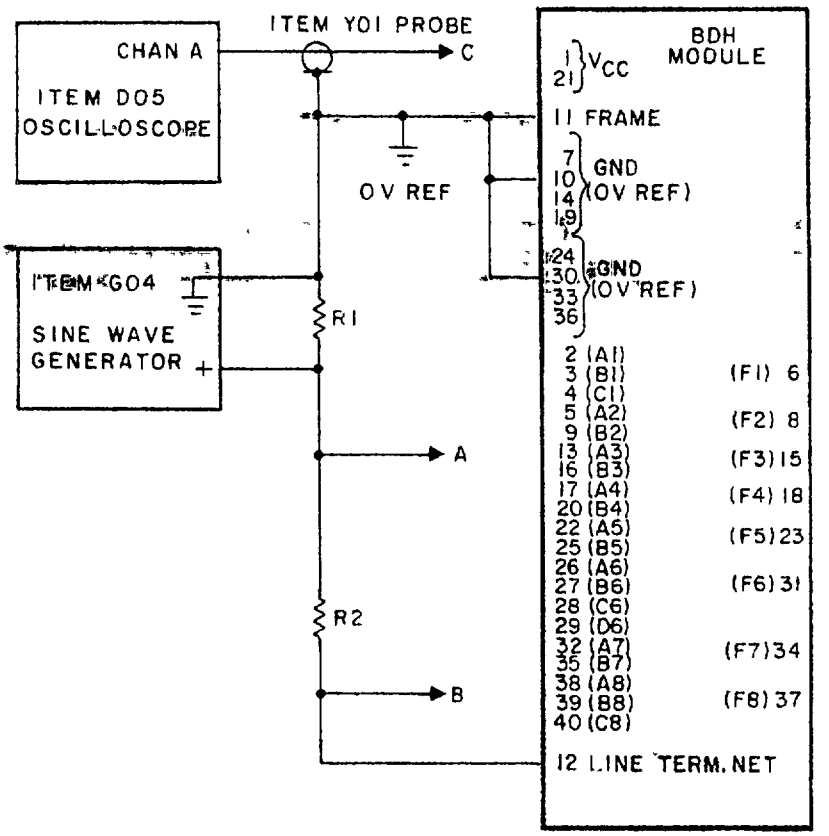
TABLE XVI Power supply transient rise time test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient rise time	10	140	µs	18,19

TABLE XVII. Parallel operation output voltage (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	Parallel operation output voltage, high level	2,3,4,5, 9,13,16	6,8,15	6,8,15	3.8	5 5	V	20
V2	Parallel operation output voltage, high level	17,20,22, 25,26,27, 28,29	18,23,31	18,23,31	3 8	5.5		
V3	Parallel operation output voltage, high level	26,27,28, 29,32,35, 38,39,40	31,34,37	31,34,37	3 8	5.5		
W1	Parallel operation output voltage, low level	2,3,4,5, 9,13,16	6,8,15	6,8,15	0 00	0.40		
W2	Parallel operation output voltage, low level	17,20,22, 25,26,27, 28,29	18,23,31	18,23,31	0 00	0 40		
W3	Parallel operation output voltage, low level	26,27,28, 29,32,35, 38,39,40	31,34,37	31,34,37	0 00	0.40		

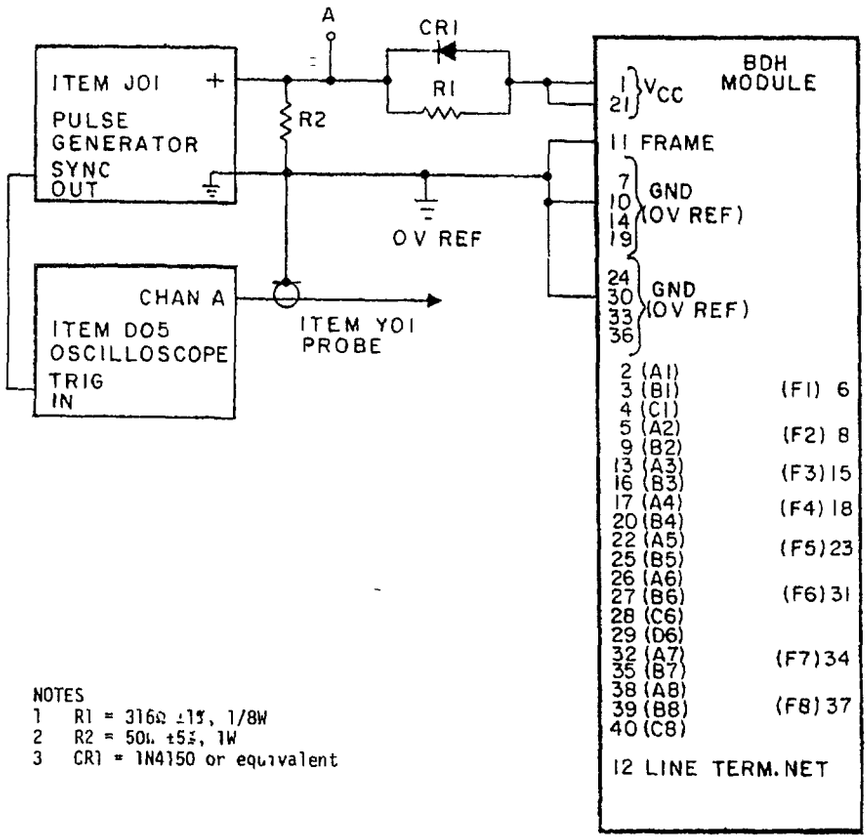
Source: https://assist.dla.mil 40Downloaded: 2026-10-20T16:08Z
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NOTES
1 R1 = 50Ω ±5%, 1W
2 R2 = 100Ω ±1%, 1/8W
3 Lead length from A to B, from B to pin 12, and from ground point of item Y01 to pin 10, each shall not exceed 2 inches (50.8 mm) maximum.
4 Test fixture capacitance (includes stray and probe at 10 MHz ±10% with the module removed from test connector)
  (a) Points A to B = 10 pF maximum
  (b) Point B to pin 10 = 12 pF maximum
5 Item G04 output at A with module connected (test condition S1)
  (a) Sine wave amplitude = 2.0 V p-p
  (b) Time per hertz = 100 ns ±10%
6 Item G04 output at A with module connected (test condition S2)
  (a) Sine wave amplitude = 2.0 V p-p
  (b) Time per hertz = 1.0 µs ±10%

FIGURE 17. Terminator amplitude (10 MHz and 1 MHz) test circuit for table XV

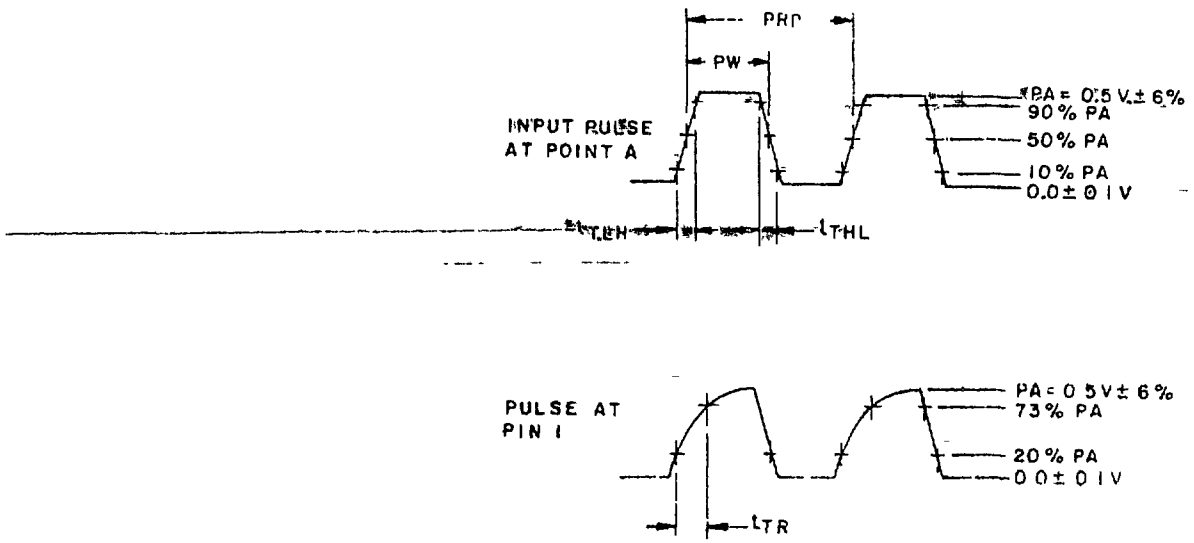
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NOTES
1 R1 = 316Ω ±1%, 1/8W
2 R2 = 50Ω ±5%, 1W
3 CR1 = 1N4150 or equivalent

FIGURE 18 Power supply transient rise time test circuit for table XVI

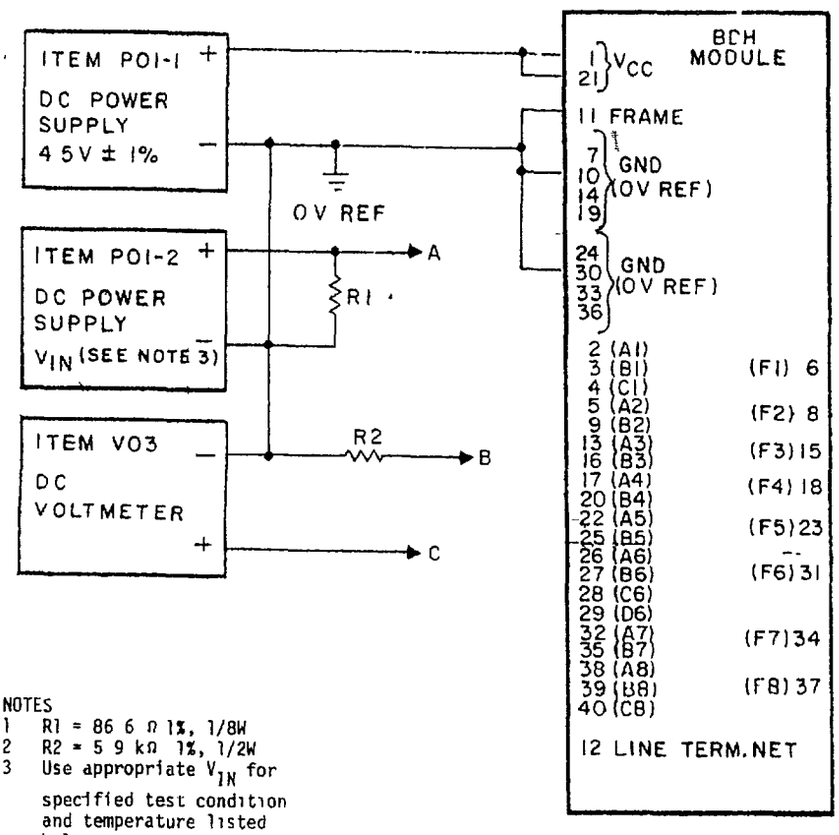
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NOTES
1 Input pulse parameters
$t_{TLH} = t_{THL}$ = 50 ns ±10%
PW = 750 µs (at 50% amplitude; ±10%
PRP = 1.5 ms
2 Measure transient rise time at pin 1 as shown

FIGURE 19 Power supply transient response test circuit and waveforms for test IV

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NOTES
1 R1 = 86 6 Ω 1%, 1/8W
2 R2 = 5 9 kΩ 1%, 1/2W
3 Use appropriate V_{IN} for specified test condition and temperature listed below

Test condition	Temperature °C	V_{IN} ±1%
U1 - U3	0	1 20
	25	1 10
	60	0 95
V1 - V3	0	2 0
	25	1 9
	60	1 8

The temperature of 25°C refers to ambient temperature 0° and 60°C refer to operating temperature

FIGURE 20 Parallel operation output voltage (high-level and low-level) test circuit for table XVII

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